Chapter test

39 Uses of microorganisms

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following are the uses of microorganisms?

   (1) production of soy sauce
   (2) production of enzymes for biological washing powder
   (3) production of human insulin

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

★ 2. Which of the following correctly states the function of the water jacket of a fermenter?

   A. To insulate the fermenter.
   B. To maintain the pressure inside the fermenter.
   C. To monitor the temperature inside the fermenter.
   D. To remove the heat produced during fermentation. ☐

3. Which of the following statements about beer-brewing is ***incorrect***?

   A. Barley grains are the raw material of beer.
   B. During malting, starch is broken down into simple sugars.
   C. Yeasts are added during the process of malting.
   D. The bitter flavour of beer comes from hops. ☐

4. During yogurt production, the pH of the mixture decreases. What causes this change in pH?

   A. release of carbon dioxide
   B. production of ethanol
   C. breakdown of lactose
   D. coagulation of milk proteins ☐

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5. Which substance produced by the yeasts causes the bread dough to rise?

   A. ethanol
   B. carbon dioxide
   C. lactic acid
   D. oxygen ☐

6. Which of the following comparisons between vaccines containing weakened pathogens and vaccines containing antigenic proteins is correct?

	Vaccines containing weakened pathogens	***Vaccines containing antigenic proteins***
A.	do not contain antigens	contain antigens
B.	pathogens may become active again in the recipient's body	antigenic proteins will not trigger immune response in the recipient's body
C.	prevent bacterial diseases	prevent viral diseases
D.	produced by culturing microorganisms	produced by genetic engineering

☐

7. Which of the following statements about antibiotics is ***incorrect***?

   A. Antibiotics are chemicals produced by microorganisms to reduce competition.
   B. Penicillin is produced by a genus of fungus called *Penicillium*.
   C. Antibiotics can be used to prevent viral diseases.
   D. Mass production of penicillin involves the inoculation of spores of the selected fungus in a fermenter. ☐

8. How does pectinase help in fruit juice production?

   A. Pectinase increases the protein content of the fruit juice.
   B. Pectinase increases the amount of the fruit juice.
   C. Pectinase breaks down the starch stored in the fruit cells.
   D. Pectinase eliminates toxins from the fruit juice. ☐

★ 9. Which of the following statements about biogas is correct?

   A. Archaebacteria are involved in biogas production.
   B. Biogas is produced by aerobic decomposition of organic matter.
   C. The major component of biogas is hydrogen gas.
   D. Biogas is a non-renewable energy source. ☐

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10. Which of the following are the advantages of using genetically modified (GM) bacteria to produce human insulin?

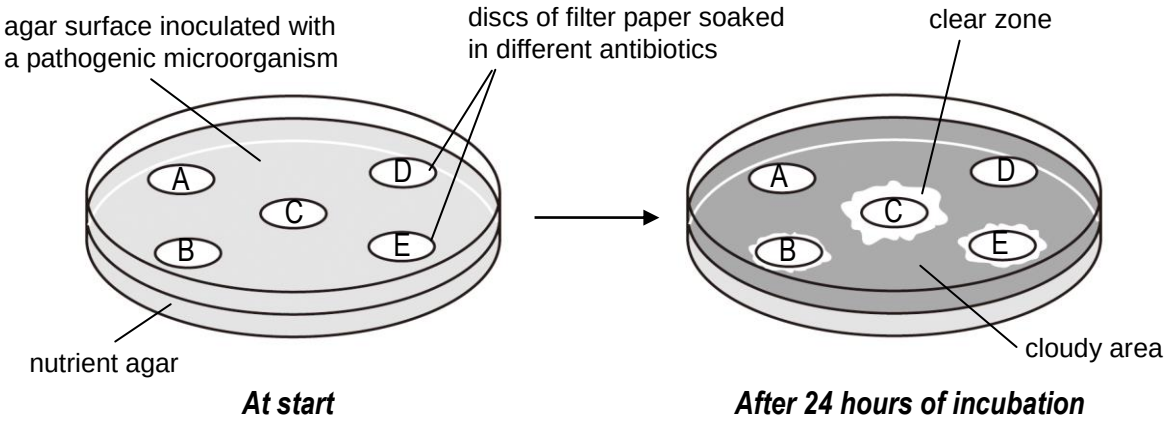
(1) The human insulin produced by GM bacteria is easier to purify.
(2) The human insulin produced by GM bacteria has a lower chance of causing allergy.
(3) The production cost is low.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	39.1	39.1	39.2	39.2	39.2	39.3	39.3	39.4	39.5	39.7

B. Problem solving (20 marks)

11. The diagram below shows an experimental set-up used to study the effects of five different antibiotics on the growth of a pathogenic microorganism.



(a) Give an example of antibiotics. (1 mark)

(b) Suggest a pathogenic microorganism that could be used for this experiment. Explain your answer. (2 marks)

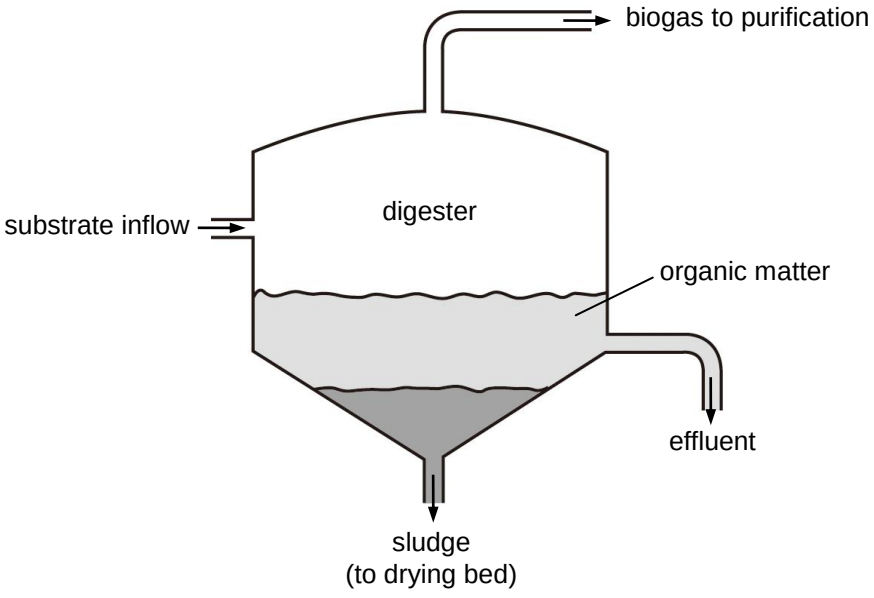
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(c) Explain the cloudy appearance on the surface of nutrient agar after 24 hours of incubation. (2 marks)

(d) Based on the experimental results, which antibiotic is the most effective against this pathogenic microorganism? Explain your answer. (5 marks)

(Total: 10 marks)

★ 12. The diagram below shows a digester used for the mass production of biogas.



(a) Suggest a source of organic matter that can be used for the large-scale production of biogas. Explain your answer. (2 marks)

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(b) What is the most important condition for the fermentation of organic matter to occur? Describe how this condition is provided in the digester. (2 marks)

(c) Name the group of microorganisms used in biogas production. Which domain and kingdom does this group of microorganisms belong to? (3 marks)

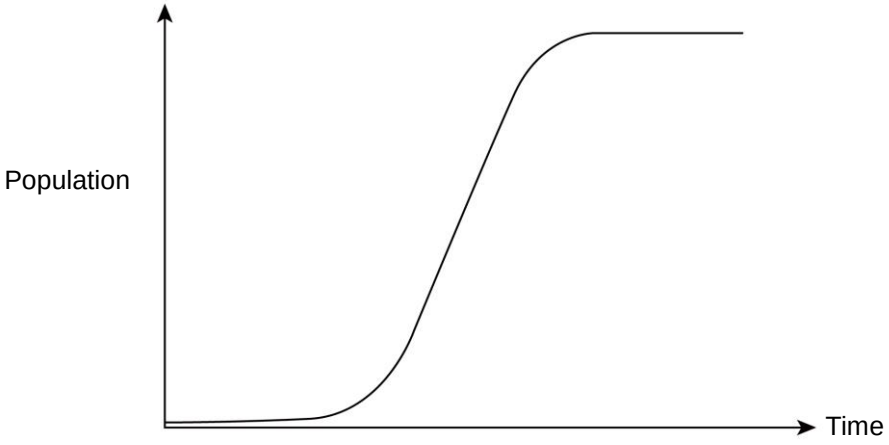
(d) (i) What is the major component of biogas? (1 mark)

(ii) Give ***two*** advantages of using biogas as a fuel. (2 marks)

(Total: 10 marks)

C. Exam challenge (10 marks)

★ 13. The graph below shows the changes in a population of microorganism present in the fermenter during beer-brewing over time.



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(a) Name the microorganism used in beer-brewing. (1 mark)

(b) What is the main raw material of beer? Name the carbohydrate that is present in a large amount in this raw material. (2 marks)

(c) Write a simple word equation to represent the fermentation reaction carried out by the microorganisms during beer-brewing. (1 mark)

(d) Based on your answers to (b) and (c), describe how to prepare the raw material so that fermentation can occur. Why is this process necessary for fermentation? (4 marks)

(e) Suggest ***two*** reasons why the population of the microorganism levels off during the later stage of beer-brewing. (2 marks)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	D	2.	D	3.	C	4.	C	5.	B
6.	D	7.	C	8.	B	9.	A	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a) Penicillin (1)
(Or other acceptable answers)

(b) Bacteria that cause gastrointestinal diseases / cholera / tuberculosis
(Or other acceptable answers) (1)
Antibiotics are effective against bacteria. (1)

(c) With the supply of nutrients in the agar plate and under suitable temperatures, the pathogenic microorganism / bacterium multiplies rapidly. (1)
Each microorganism / bacterium grows into a colony which is cloudy in appearance. (1)

(d) Antibiotic C (1)
The clear zone around the paper disc with antibiotic C is the largest, (1)
The antibiotics diffuse out from the paper discs. (1)
The further away from the paper disc, the lower is the concentration of antibiotic. (1)
A more effective antibiotic can inhibit bacterial growth even at a lower concentration. (1)

(Total: 10 marks)

12. (a) Faeces / sewage sludge / food leftovers from kitchens (1)
They are produced every day in large quantities. (1)

(b) Anaerobic condition / absence of oxygen (1)
The digester is a closed system. (1)

(c) Methanogens (1)
This group of microorganisms belongs to the domain Archaea (1)
and the kingdom Archaebacteria. (1)

(d) (i) Methane (1)

(ii) Biogas is less-polluting than fossil fuels as its combustion produces fewer air pollutants. (1)
The raw materials used to produce biogas are renewable. (1)
(Or other acceptable answers)

(Total: 10 marks)

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C. Exam challenge (10 marks)

13. (a) Yeasts (1)

(b) Raw material: barley grains (1)

Carbohydrate: starch (1)

(c) Simple sugars → ethanol + carbon dioxide (1)

(d) Soak the barley grains in water to induce germination. (1)

This activates the production of enzymes in barley grains. (1)

The enzymes catalyze the breakdown of starch / food reserve into simple sugars. (1)

This provides the sugar source for the subsequent growth and fermentation of yeasts. (1)

(e) Ethanol accumulates to a high concentration which is toxic to yeasts. (1)

Sugars start to deplete and the yeasts compete for food. (1)

(Or other acceptable answers)

(Total: 10 marks)

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